

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0922

Us Ord Depot-Herlong 138 kV
February 24, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0922. The Interconnection Customer proposes a 121.2 MW solar photovoltaic facility interconnecting at the Us Ord Depot-Herlong 138 kV substation in the Virginia area. The request was received on January 15, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,545,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LOWMOOR 345-138 kV Autotransformer	ASPEN - LOWMOOR 345 kV	113.5	449	8.6
S. LAND PARK & 35TH AVE - PRINGLE 138 kV	MANATEE - NEW HOPE AP 345 kV Line	108.8	216	6.7
US ORD DEPOT-HERLONG - DUPLESSIS 138 kV	KEAN MTN AND GRASSY CRK - BARR LAKE 345 kV Line	106.9	203	40.1
DUPLESSIS - HYAK 138 kV L	BARR LAKE - KEAN MTN AND GRASSY CREEK 345 kV	106.7	198	40.1
RED HILLS - VASEAUX LAKE 138 kV	HNTON AP - LOWMOOR 345 kV	106.2	415	8.2
MCCLAIN ENERGY FACILITY - VASEAUX LAKE 138 kV	LOWMOOR - HINTON AP 345 kV	106.2	418	8.2
FRIENDSHIP - HYAK 138 kV	KEAN MTN AND GRASSY CREEK - BARR LAKE 345 kV line	102.8	200	26.6
BEARDSLEY ST - RED HILLS 138 kV	LOWMOOR - HINTON AP 345 kV	102.0	423	8.2
OROVADA - CARLTON HILLS 138 kV	ATOKA - CARLTON HILLS 345 kV	101.4	282	33.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOWMOOR 345-138 kV XFMR	Replace transformer with higher-rated unit	\$15,715,000	9% (\$1,361,000)
S. LAND PARK & 35TH AVE - PRINGLE 138 kV	Reconductor line	\$1,728,000	8% (\$139,000)
DUPLESSIS SUBSTATION - US ORD DEPOT-HERLONG 138 kV	Reconductor line	\$2,752,000	50% (\$1,385,000)
DUPLESSIS - HYAK 138 kV	Reconductor line	\$1,092,000	50% (\$550,000)
RED HILLS - VASEAUX LAKE 138 kV	Reconductor line	\$6,129,000	7% (\$409,000)
MCCLAIN ENERGY FACILITY - VASEAUX LAKE 138 kV	Reconductor line	\$3,889,000	7% (\$260,000)
FRIENDSHIP - HYAK 138 kV Line	Reconductor line	\$1,436,000	74% (\$1,060,000)
BEARDSLEY ST - RED HILLS 138 kV	Reconductor line	\$5,067,000	7% (\$338,000)
OROVADA - CARLTON HILLS 138 kV	Reconductor line	\$182,000	24% (\$43,000)
	Total allocated to project		\$5,545,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.